

Effect of ergonomic aspects on single- and multiproduct assembly-line balancing problems

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Abstract

Incorporating ergonomic considerations into an assembly-line balancing problem (ALBP) enhances productivity and minimizes ergonomic concerns. The assembly process, characterized by repetitive motions and handling numerous components, can lead to worker overload. Consequently, the inclusion of ergonomic aspects results in an appropriate distribution of assembly operations and relative workloads. This study investigates a multi-objective ALBP aimed at minimizing the number of workstations, overall skill level required, and variance in workers' energy expenditure across workstations. To address the ALBP while considering the ergonomic aspects, this study proposes an approach based on the non-dominated sorting genetic algorithm II (NSGA-II) and multi-objective simulated annealing (MOSA) using Pareto optimality. A comparative analysis of the NSGA-II and MOSA is conducted in single- and multiproduct production scenarios, and a computational study involving various factors is performed to identify the dominant algorithm. The computational analysis indicates that the runtime performance of MOSA is 73.287% better than that of NSGA-II; therefore, MOSA outperforms NSGA-II. This study aims at applying scientific knowledge concerning manufacturing ergonomics to assist manufacturing industries in enhancing their productivity.

KEYWORDS

assembly line balancing, cognitive abilities, cognitive skills, ergonomics, multi-objective optimization, multi-objective simulated annealing, non-dominated sorting genetic algorithm II

1 | INTRODUCTION

Ergonomics is an important aspect to consider when solving problems related to manual assembly because of its effects on productivity and ergonomics-related issues. Frequent execution of repetitive movements and handling of components are among the main factors that characterize the assembly process, potentially resulting in worker overload. An appropriate distribution of assembly

operations based on workers' skills reduces workload and provides a cognitively and ergonomically friendly work environment.

Businesses now understand the value of effective production systems in the face of the fierce competition between developing and changing industries. In such assembly systems, lowering manufacturing costs and achieving customer demand are possible only with modifications to traditional production systems. In an assembly system, unidentified problems can create issues in finished

Relevance to industry: Ergonomic aspects in an assembly line balancing problem helps in improving productivity and minimizing human errors. The proposed algorithms in this research are well suited to industries where assembly activities necessitate dexterity and manual skills gained through experience. In such production units, skilled workers are a valuable asset like any other resource. These algorithms can design assembly plans based on specific individuals available to the organization, saving workers from being overloaded and stressed. The inter- and cross-disciplinary character of this study enables for a wide range of human resource management concerns to be addressed in manufacturing unit. Selection training and education, labor-management interactions, human skills development, and knowledge-enhancing technology are all examples of specialized subject areas of interest.

products. For example, in the automobile industry, if an O-ring or small part is misplaced in a subassembly by a worker, the entire functionality is affected. The challenge was to identify what went wrong during the assembly process. Kosina (2015) explains the Shainin methodology, which leads companies to use the automotive sector as an effective problem-solving tool. The Shainin method aids industries in assembling barrels to assemble a full diesel fuel-injection system.

Ongoing progress in the understanding of human cognitive skills requires new data acquisition, synthesis, and integration through studies and laboratories. Giridhar and Panicker (2023) conducted a case study in which workers were assigned to specific workstations based on their cognitive skills, and their results demonstrated an enhancement in the productivity of the entire production unit. Worker allocation in an assembly line was done based on cognitive aspects of information and material presentation matters. The experimental methodology analyzed the skill level of the worker and the skills required for each task, providing input for this study. This study presents a generalized version of worker allocation and verifies the objective and subjective assessments using mathematical proofs. This is a case study of a small-scale industry in southern India. This study considered the problem of 10 tasks and 6 workers for a single-product assembly line and six products for a multiproduct assembly line. The inputs from a real-time industry are the precedence relationship, constraints, and objective functions for a single-product and multiproduct assembly line.

In several developed countries, employees' mental health and wellbeing are addressed by identifying the appropriate skills. In developing countries, such as India, more consideration needs to be given to the right skills and workplace for workers with specific skills; instead, workers are randomly allocated. Therefore, this study formulated a mathematical model with inputs from a small-scale industrial unit to address this problem. Although many studies have focused on physical ergonomics and workstation design, cognitive aspects require attention. In manual assembly lines, mental stress decreases productivity because of the repetitive and unfavorable working conditions. Thus, cognitive skill assessment helps place them at the correct workstation. This research proposes a systematic approach to allocating workers to the workstations at which they belong, while considering ergonomic issues that may improve productivity, workers' working conditions, occupational health, and safety, and, indirectly, cost-effectiveness.

Assembly-line balancing (ALB) is performed by assigning tasks to various workstations in the production system. These tasks involve fundamental activities to convert raw materials into finished products. Line balancing is a well-known optimization technique used in production systems. A mass-production system assembles identical or interchangeable parts or components into a final product in multiple phases at various workstations (Kumar & Mahto, 2013). The implementation of line-balancing techniques must be refined with the advent of digital transformation. The assembly-line balancing problem (ALBP) is concerned with the optimal assignment of various assembly operations to production

line workstations. The purpose of the ALBP is to optimally assign various operations in an assembly process to production line workstations. Dolgui and Proth (2010) discussed the economic impact of ALBP on the design of linear production systems. Researchers have developed different algorithms and programming techniques to reduce complexity in this context. This approach is essential for solving problems for a firm to economically maintain an assembly line's productivity and flexibility. Battini et al. (2016) introduced a new multi-objective model to solve ALBPs based on a recently developed predetermined motion energy system technique to rapidly estimate energy expenditure values. A real-world case study with a simple numerical example analyzed Pareto frontier behavior and identified variations in parameters related to energy and time values.

Mixed-product assembly lines have ergonomic concerns that can reduce worker and line productivity. Bautista et al. (2016) introduced an approach for balancing assembly lines that simultaneously considers temporal, spatial, and ergonomic factors. They conducted a case study with Nissan using several mathematical models developed to study its behavior. Assembly lines are mass-production systems that manufacture similar products in large volumes at a low cost. Manufacturers utilize mixed-model assembly lines to generate customized and nonidentical items in response to customer demands. Battini et al. (2017) prevented ergonomic risks by using an integrated plan to solve the ALBP. Furthermore, Alghazi and Kurz (2018) suggested mixed-model line-balancing integer programming to reduce the number of workers, considering workers, zoning, task assignment, and ergonomic limitations. Only a few studies have explored component picking at assembly lines and its significance in the ALBP. Bortolini et al. (2017) proposed a unique multi-objective optimization problem for assigning assembly tasks to stations by distinguishing the assembly activities involved in task execution and component picking.

This study addresses the multi-objective optimization of the ALBP, incorporates ergonomic considerations in designing production systems, and explores the applicability of the Pareto optimality concept. To this end, this study proposes an original approach based on an extension of two leading meta-heuristic algorithms: the genetic algorithm and simulated annealing. The proposed approach significantly contributes to ergonomics in assembly line research by applying Pareto optimality over a weighted approach, as reported in the literature.

The proposed approach uses the following two algorithms to solve the ALBP for single- and multiproduct models:

- Nondominated sorting genetic algorithm II (NSGA-II)
- Multi-objective simulated annealing (MOSA)

Finally, this study evaluated the performance of the NSGA-II against the MOSA. This study explores the optimization of assembly line ergonomics, considering a broad spectrum of ergonomic issues, with a focus on the operation and management of manufacturing systems in shop-floor environments.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and provides the rationale for this study. Section 3 discusses the modeling methodology for the ALBP. Section 4 proposes an approach to solving the ALBP, and Section 5 presents the results and discussion. Section 6 presents a comparative analysis of NSGA-II and the MOSA used in the proposed approach. Section 7 discusses the findings of the study and their industrial implications. Finally, Section 8 presents the conclusions and suggests directions for future work.

2 | LITERATURE REVIEW

A review of the related literature was conducted and categorized as follows:

- Relevance of ergonomics in assembly lines
- ALB
- Application of the following algorithms for similar optimization problems
 - NSGA-II
 - MOSA

2.1 | Relevance of ergonomics in assembly lines

This subsection briefly introduces the concepts of ergonomics and the related research. Ergonomics is a branch of science that explains the scientific analysis of working conditions. It focuses on the physical and psychological features of human operators and their talents, constraints, and demands. Researchers have conducted several studies to identify worker issues in the assembly lines of large factories operating with modern convenience. Psychologist Murrell introduced the contemporary meaning of ergonomics and founded the first national ergonomic society to promote worker health, safety, and wellbeing. According to Naeini and Mosaddad (2013), ergonomics aims to provide a workstation that balances worker safety and productivity to protect workers' health and physical and psychological wellbeing. Manual assembly is characterized by the frequent execution of repetitive and tedious tasks and physical stress.

The primary goal of any manufacturing unit is to increase profits; traditional assembly line optimizations only consider economic factors, such as the production rate. Otto and Scholl (2011) introduced ergonomic risks into ALB, in which the indirect costs incurred owing to worker health issues, such as medical costs and absenteeism, are considered. If a skilled worker is absent, a replacement is required to run the production line; however, productivity is reduced because of the new worker's learning process and slow performance. The physical capability of a worker is influenced by various factors, including sex, age, genetic characteristics, personal lifestyle, and workplace layout (Michalos et al., 2018). Physical capabilities decreased over time for all workers, particularly

those who were older. Although the physical capabilities of experienced workers are reduced, their professional skills remain an asset for the production units. Older workers will become an increasingly important workforce element as the population ages.

Iranzo et al. (2020) used an exoskeleton at an automotive assembly plant. The experimental results showed an apparent reduction in the low and medium activities of the muscles responsible for arm flexion when using an exoskeleton. Similarly, Guimarães et al. (2014) reported a macro-ergonomics intervention at a footwear company in Brazil and its evaluation by workers. The findings of this study indicate that participatory ergonomics has the potential to identify issues and propose solutions that are more readily embraced by workers compared to managers. Furthermore, it suggests that a macro ergonomic intervention, with a primary emphasis on work organization, can yield favorable outcomes in terms of personnel, health, and production, even in the face of management's reluctance to embrace change.

Village et al. (2017) explored the ergonomics of assembly line manufacturing and the limitations of implementing them to improve assembly quality. A scorecard and rating system was created as part of this research. Two ergonomic approaches were developed to improve the quality throughout the early stages of design and assembly line ramp-up. Moreover, de Negreiros et al. (2019) investigated the breaks taken by workers during their shift hours in cell and serial assembly layouts, which influenced the layout of workers' health. Biomechanical exposure analyses and questionnaires were used to examine the physical, cognitive, and mental health of workers in the automobile industry, considering the cycle and break times. It has been reported that rest breaks in the workplace, when performed at appropriate intervals, can improve the general health and productivity of workers, lower healthcare expenses, and increase industrial production. Caputo et al. (2019) designed a framework based on a concurrent engineering method that utilized the digital twins of the Fiat Chrysler automobile assembly line. This framework helps reveal ergonomic problems in an assembly line, and design issues are solved during the design phase rather than the production phase. With this method, significant cost savings for the repair of design flaws can be realized, as well as a significant reduction in the time to market.

Based on these studies, it becomes evident that ergonomic considerations are essential when treating ALBP. Nonetheless, it is the ethical responsibility of every employer to simultaneously protect employee health and wellbeing and improve production rates.

2.2 | Research in ALB

This section provides a brief review of research on ALB. This topic has attracted significant research attention since Salvesson (1955) proposed a mathematical formulation of an ALB model. For a Simple Assembly-Line Balance Problem (SALBP), heuristic solution procedures were developed and recorded in a review by Scholl and Becker (2006).

An ALB assigns tasks to an assembly line to achieve the desired production rate and results. The tasks are assigned in an orderly sequence of workstations, with task precedence constraints satisfied and performance metrics optimized. The challenges in ALB are primarily concerned with properly distributing tasks to workstations in a balanced manner while considering the precedence relationship and optimizing the number of workstations or the cycle time. Assembly line construction is a strategic and high-cost choice that is critical for properly designing an assembly line and balancing the workload at workstations. The advantages of line balancing include enhanced efficiency and productivity in the manufacturing units. Eghtesadifard et al. (2020) explained that when the number of workstations is optimized for a given cycle time in the classical model of ALB, it is known as an SALBP of type 1 (SALBP-1), whereas minimizing the cycle time for a given number of workstations is known as an SALBP of type 2 (SALBP-2).

The gap between the SALBP-1 model and the planning of real-time assembly line configurations with all the constraints is vast. Mass producers utilize assembly lines in various industries, including the production of various products such as electronic goods, automobiles, and airplanes (Boysen et al., 2008). Automobile assemblies are undoubtedly the most widely used components. Hundreds of employees must assemble thousands of parts and coordinate with machines at several workstations on a typical automotive assembly line, with a cycle time ranging from 60 to 90 s (Battini et al., 2013).

To account for the constraints related to the problem parameters, Ozdemir et al. (2021) introduced a fuzzy multi-objective problem for ALB, which was evaluated using robust optimization approaches and ergonomic risks. Pilati et al. (2020) introduced the ALBP as the best method to assign numerous operations, leading to various assembly line processes. Weckenborg and Spengler (2019) considered an ALBP for cost-efficient system configurations, in which robots and human workers could be allocated to a production line. Boysen et al. (2022) reviewed previous studies on ALB published in 2007. This review reports a methodology for retrieving precedence data, reviewing new problem variants, and developing models of ALB extensions as adopted in the literature.

Anel et al. (2022) investigated a mixed-model ALBP type 2 with single-sided linear assembly line configurations. These are the typical configurations adopted in small- and medium-sized industrial environments. A genetic algorithm (GA)-based solution was proposed to resolve the balancing operation times and achieve algorithmic transparency (AT). The AT created using the new matrix methodology is validated using a real-world paradigmatic example. Albus and Seeber (2021) introduced a mixed-integer programming model to solve the dynamic resource-allocation problem for assembly lines, which was modeled as an ALB.

According to the literature review, ALBPs are categorized based on their objective functions and problem structures. Based on the objective functions, ALBPs are named type F, type E, and types 1, 2, 3, 4, and 5. Similarly, based on the problem structure, ALB models are classified as single-model ALBPs or multi-model (mixed model)

ALBPs. This study focuses on both classifications. However, this study addressed an ALBP type F from the objective function category, which is a feasibility-based problem, to find a viable solution for a given number of workstations and cycle times. Our study implemented an ALBP solution that minimized the number of workstations, the total skill level required of assigned workers, and the energy expenditure variance of workers among workstations. For simplicity, the objective of minimizing the number of workstations was relaxed.

In summary, this study reports a combination of ALBP variants, namely, the type-F problem and ALBPs based on product structures.

2.3 | Application of MOSA as a solution to a similar optimization problem

This subsection summarizes the application of the MOSA algorithm. Varadharajan and Rajendran (2005) explored the problem of permutation flow shop scheduling to minimize the makespan and total flow time of jobs. MOSA was proposed as a solution, wherein two initial sequences were obtained using local search heuristics. A comparative study was presented between MOSA and multi-objective genetic local search (MOGLS), an elitist, non-dominated sorting GA, and gradual priority weighting. Finally, it was reported that the proposed MOSA required less computational effort.

Khan and Govindan (2011) conducted a study to minimize the objective functions, makespan, and maximum tardiness of jobs, which was considered a multi-objective permutation flow shop problem. They developed and evaluated the MOSA with random problems. The efficiency of MOSA was compared with that of different multi-objective genetic algorithms, such as the strength Pareto evolutionary algorithm II (SPEA-II) and MOGLS. According to the computational results, MOSA outperformed SPEA-II and MOGLS.

Considering the computational efficiency of MOSA, most researchers have adopted this approach to solve problems in different domains. Vieira and Ribas (2008) demonstrated how fractional factorial analysis could be applied to the configuration of simulated annealing to optimize master production scheduling problems. Cakir et al. (2011) applied the MOSA technique to solve ALBPs at parallel workstations. Lara et al. (2018) worked on a multi-objective meta-heuristic algorithm based on simulated annealing for political districts, and a comparison study was performed using 12 real datasets. Similarly, Dinh et al. (2019) explained the ALBP in the garment industry using simulated annealing.

Fang et al. (2020) considered the design and balancing of mixed-model disassembly lines using multi-robotic workstations. A mixed-integer evolutionary simulated annealing algorithm was developed to minimize cycle time, peak workstation energy consumption, and total energy consumption. Computational experiments suggested that the MOSA outperformed other multi-objective algorithms in terms of solution quality and computational efficiency.

Therefore, as per the review of related literature, MOSA has been reported as a computationally efficient approach to solving

combinatorial optimization problems. The MOSA can be applied to construct Pareto-optimal solutions that are non-dominated in the explored solution space.

2.4 | Application of NSGA-II as a solution to a similar optimization problem

A brief review of earlier attempts to generate non-dominated solutions using a non-dominated sorting GA is presented in this subsection.

Murata et al. (1996) applied a multi-objective genetic algorithm (MOGA) to a bi-objective flow shop scheduling problem to determine nondominated solutions. Furthermore, the model was extended to address multiple objectives. Interestingly, Ishibuchi and Murata (1998) extended MOGA by incorporating a local search procedure into the offspring, and the algorithm was termed MOGLS. Srinivas and Deb (1994) proposed a non-dominated sorting GA. This approach based on GA Bagchi (1999). Chang et al. (2002) proposed a gradual priority weighting (GPW) approach-based GA.

Multi-objective evolutionary algorithms (EAs) that adopt non-dominated sorting and sharing have drawbacks such as computational complexity, non-elitism, and the requirement to define a sharing parameter. Deb et al. (2002) suggested a non-dominated sorting-based multi-objective EA (MOEA) called NSGA-II, which alleviates all three drawbacks.

According to Yusoff et al. (2011), the computational performance of an algorithm depends on the optimization approach adopted within the algorithm. Consequently, several researchers have proposed improvements to the NSGA-II algorithm to address some of its flaws while solving optimization problems. Rajkumar et al. (2014) stated that NSGA-II is a fast MOEA that uses an elitism-preserving approach to implement elitism in multi-objective searches. The quality of convergence toward a genuine Pareto-optimal set is improved by elitism.

NSGA-II is an upgraded version of Deb's non-dominating sorting GA, first described in 2001 (Wang, 2016). Over the years, NSGA-II has proven to be a very good MOGA used by researchers to solve many multi-objective optimization models. The ALBP is a multi-objective optimization problem. Researchers have applied various methods to handle multiple objectives, such as the weighting method (Aydemir-Karadag & Turkbey, 2013; Dalle Mura & Dini, 2019), lexicographic order (Kalayci et al., 2015), and Pareto dominance (Ding et al., 2010). Although the weighing procedure is simple, it requires the establishment of appropriate weights. In a multi-objective optimization problem, the goodness of a solution is determined by its dominance. Using the Pareto dominance method, the non-dominated set of the entire feasible decision space, called the Pareto-optimal, is determined, and a Pareto Front is plotted. If no objective can be improved without compromising at least one other objective, the Pareto Front provides a set of non-dominated solutions that can be chosen as optimal.

Parizad and Hatziaodoniu (2020) presented a security/stability-based Pareto-optimal solution for the optimal placement and sizing of renewable energy generation. By treating the function as a bi-objective problem and ignoring one objective in each run, NSGA-II obtains a Pareto-optimal front for solving three-objective optimization issues. The literature shows that NSGA-II can yield a better Pareto front in multi-objective optimization problems. Moreover, a dimensionality-reduction approach can be applied to simplify the computational process by fixing one of the objectives among the multiple objectives considered.

2.5 | Cognitive skills

Heimerl and Kolisch (2010) proposed a nonlinear continuous optimization model to solve the problem of worker assignments by incorporating learning, knowledge depreciation, and company skill-level goals. Liu et al. (2021) addressed multiskilled worker assignment and ALBP by considering energy consumption. Blum and Miralles (2011) studied the ALBP with operation and multiskilled workers assigned to workstations. Similarly, Zaman et al. (2012) studied an ALBP with multiskilled workers to minimize the weighted sum of the cycle time and total idle time. A genetic algorithm-based heuristic was proposed to solve this problem based on these characteristics. Oksuz et al. (2017) investigated a problem with a U-shaped assembly line, considering multiskilled workers and worker performance. A mixed-integer linear programming model was formulated to maximize the line efficiency. Recently, Giridhar and Panicker (2022) investigated how various cognitive abilities influence the overall cognitive skills of the subjects. Cognitive neuroscientists increasingly agree that the ongoing progress in understanding human cognitive skills requires new data acquisition, synthesis, and integration through studies and laboratories. Efforts to focus more on cognitive skills, effective ways of identifying tasks, and experimentation using a cognitive atlas have contributed to the study's experimental design. Previous studies have considered skills in real-time; however, some authors still need to conduct experimental justifications or real-time comparisons.

2.6 | Ergonomics and its role in assembly system design

Assembly system design (ASD) focuses on balancing and sequencing procedures, and solves the ALBP in different layout configurations by developing exact or heuristic solutions. Gokçen and Agpak (2001) explained the classic ALBP, which includes assigning assembly tasks with a certain processing time and considering precedence relations to a workstation, which helps optimize several objectives. Baykasoglu et al. (2017) reviewed ASD studies from the initial stages and explained ALBPs types 1, 2, 3, 4, E, and F. In this study, a systematic approach to handling ASD, which consists of three subproblems while considering ergonomic risk factors, was considered. The ASD

problem deals with subassembly, main assembly, and layout problems, considering ergonomic risk factors. The OCRA (Occhipinti, 1998) was used to perform ergonomic risk assessments. Solving this problem based on a real-life case study that analyzes the upper limbs of the body, such as the hands, wrists, elbows, and shoulders, and performs ergonomic analysis for each part of the body separately (left and right), is a good contribution to physical ergonomic research.

Problems can be divided into the following three categories, depending on the number and variety of products:

- (i) A single-model line is a classical configuration in which a single model of a unique product is produced.
- (ii) A mixed-model line is one wherein several variants of a basic product, referred to as models, are simultaneously produced (Akpınar & Baykasoglu, 2014a, 2014b).
- (iii) A multi-model line in which different models with significant differences were produced.

An assembly line worker assignment and balancing problem (ALWABP) occurs in many assembly lines when the operation time for each task differs according to the worker. The operation times of the tasks vary because of the workers, and the problem requires a simultaneous solution to the double-assignment problem. In this study, instead of solving the problem using the conventional method, we attempted to incorporate the cognitive skills obtained. Salveson (1955) presented the first article on ALBP, and recent literature has also studied ergonomic risk assessment in assembly lines. Although many research attempts have been made on ALBP, a combination of physical and cognitive aspects with real-time examples is scarce.

Traditional procedures only consider task-time analysis and precedence relations, and human cognitive skills must often be integrated into these techniques. The workload among workers varies, and operators repetitively perform the same type of work in manual assembly lines. Consequently, repetitive work has a cumulative effect on the workers. Assembly lines that need to be better designed ergonomically cause a lack of productivity and increase the workers' cognitive workload. The results of this study suggest that identifying skills based on tasks improves worker allocation and thus motivates workers. Incorporating cognitive aspects improves process quality and operator productivity. Drury (2000) analyzed the relationship between ergonomics and quality improvements in a production system and proposed guidelines for a discipline that includes quality and ergonomics.

Carey and Gallwey (2002) explained the importance of workplace ergonomics and how it helps solve work-related musculoskeletal disorders (WMD). Several studies have considered that the links between ASD, balancing problems, and ergonomics evaluation techniques are already in practice but are still unable to solve some problems related to the cognitive skill differences among workers. Several studies have focused on physical ergonomics; however, one that includes both cognitive and physical ergonomic perspectives in real-time studies must be listed.

2.7 | Findings from the literature review

From 1990 to 2000, the focus of researchers was on SALBPs and heuristic methods; however, in the post-2000 era, the focus changed to variant ALBPs and their solutions through novel meta-heuristic algorithms (Eghtesadifard et al., 2020). The literature review revealed that researchers still focus on SALBP type 1 and type 2 problems based on the type of objective function. In recent years, numerous studies have been conducted on this topic. In addition, among multi-objective optimization algorithms, approaches based on simulated annealing and genetic algorithms are widely preferred.

Similarly, most studies on the ergonomic aspects of assembly lines have been based on physical ergonomics, whereas skill-based studies on worker performance are limited. Research on line-balancing problems has adopted a weighted-sum approach when dealing with multi-objective problems. Multiple objectives were assigned subjective weighting values to merge them into a single-objective function. The relative subjective importance of the objectives determined their weight values. A drawback of the weighted-sum method is that the solution depends on the assigned weights. A small variation in the weights can significantly alter the objective function values. In addition, it is difficult to assign appropriate weights to each factor. Thus, the present study distinguishes itself and introduces novelty. The present study does not consider the weighted-sum approach because the measures of variations in the objective function cannot be assigned certain weights. Moreover, in practical production scenarios, these weights cannot be measured because the skills, number of workstations, and variance in energy expenditure have different measurement units.

For the multi-objective optimization model, the goodness of the solution was determined by dominance. In this regard, the determination of a non-dominated set is essential. To address these gaps in the literature, this study compares two algorithms, NSGA-II and MOSA, for single- and multiproduct ALBPs using the Pareto optimality approach. This is another unique aspect of this study.

To the best of our knowledge, no study has compared solution approaches based on NSGA-II and MOSA to address single- and multiproduct (mixed-model) ALBPs with the inclusion of ergonomic considerations. Thus, this study contributes significantly to the literature, and the results demonstrate potential applications of ALBPs. Baykasoglu et al. (2017) stated that, while considering traditional objectives, one should recognize the ergonomic issues that play an important role at the workplace and system levels. It is well known that productivity can be improved by applying the principles of ergonomics in production environments, specifically ASD. The author identified lower productivity and a higher number of WMD among workers owing to the poor design of the assembly system. Çimen et al. (2022) presented a detailed review of the assembly line rebalancing problems (ALRBPs) to provide a better insight into the theoretical and practical applications of ALRBPs. A structured database search was conducted, and 41 ALRBP papers published between 2005 and 2022 were classified based on the problem structure, objective functions, problem constraints, reasons for rebalancing, solution approaches, and type of

data used for solution evaluation. Finally, future research directions were identified and recommended.

Akyol and Baykasoğlu (2019) proposed a new type of ALWABP that considered ergonomic risks. ALWABP occurs when task times vary according to the assigned worker. Although the operation time of a task is assumed to be fixed in classical assembly lines, it depends on the operator who executes the task in the ALWABP. The experimental results showed that the ergonomic environment could be improved by considering ergonomic risk factors.

3 | MODELING METHODOLOGY IN AN ALBP

The mathematical model that formed the basis of this study was obtained from the multi-objective optimization model developed by Dalle Mura and Dini (2019). The energy expenditure for various micromovements in an assembly line was computed using the empirical relationships in Garg et al. (1978).

3.1 | Mathematical model

The decision variables, objective functions, and problem constraints are described in the following subsections.

Notations applied

Decision variables

m Number of workstations

S Skill of worker assigned to n^{th} workstation and $n = 1, \dots, m$

η_n Energy gap of n^{th} workstation and $n = 1, \dots, m$

$\bar{\eta}$ Mean energy gap between workstations

Objective functions

Objective function 1

The first objective function is to minimize the number of workstations m .

$$\text{Min } O_1 = m \quad (3.1)$$

Objective function 2

The second objective function is to minimize the total skill of assigned workers.

$$\text{Min } O_2 = \sum_{n=1}^m S_n \quad (3.2)$$

Objective function 3

The third objective function minimizes the standard deviation of the energy expenditure of workers at workstations.

$$\text{Min } O_3 = \sqrt{\frac{\sum_{n=1}^m \eta_n - \eta m}{m}}, \quad (3.3)$$

where $\bar{\eta}$ is the mean of the energy gap between workstations and ηm is the energy gap of workstation m and is computed using Equation (3.4).

$$\eta m = 1 - \frac{E_j}{0.0143 \times OEL_j \times CT}, \quad (3.4)$$

where E_j is the sum of the energy required to perform all tasks by worker j at workstation m .

OEL_j is the occupational energy limit of worker j and

CT is cycle time (min).

Constraints: The above-stated objective functions are imposed with the following constraints:

- Precedence constraint
- Skill constraint
- Energy constraint
- Cycle-time constraint

The constraints are explained as follows.

Precedence constraint

Each assembly sequence followed a precedence constraint. The precedence relation matrix $p_{i,z}$ is given by Equation (3.5)

$$p_{i,z} = \begin{cases} 1 & \text{if task } i \text{ precedes task } z \\ 0 & \text{if task } i \text{ does not precede task } z \end{cases} \quad (3.5)$$

Skill constraint

The worker assigned to the workstation should have a skill (S_j) greater than or equal to the maximum skill ($S_{\max_m}$) required for the task performed at workstation m , which is given by Equation (3.6).

$$S_j \geq S_{\max_m} \quad (3.6)$$

where,

$$S_i = \begin{cases} 0, & \text{for low skill level} \\ 1, & \text{for intermediate skill level} \\ 2, & \text{for the task that requires high skill} \end{cases} \quad (3.7)$$

Energy constraint

Workers should not work beyond their physical capacity. The energy required for micro-movement n of task i by worker j (e_{nij}) was computed using the empirical relation proposed by Garg et al. (1978).

Some of the relations applied are as follows:

The energy required for arm lowering,

$$e_{nij} = 10^{-2} \times [0.093 \times WW_j \times (h2_n - 0.81) + (1.02 \times L_n + 0.37 \times g_j \times L_n) \times (h2_n - h1_n)] \quad (3.8)$$

The energy required for arm lifting,

$$e_{nij} = 10^{-2} \times [0.062 \times WW_j \times (h2_n - 0.81) + (3.19 \times L_n + 0.52 \times g_j \times L_n) \times (h2_n - h1_n)] \quad (3.9)$$

The energy required for walking,

$$e_{nij} = 10^{-2} \times (51 + 2.54 \times WW_j \times (SW_n)^2 + 0.379 \times WW_j \times GS \times SW_n) \times t_n \quad (3.10)$$

The total energy expenditure of worker j to perform micromotion n at workstation m , with a subset of $1...N_t$ tasks are given by (3.11):

$$E_j = \sum_{n=1}^{N_{mi}} \sum_{i \text{ subset of } 1, \dots, N_t} e_{nij} \quad (3.11)$$

where N_{mi} is the total number of micromovements in task i and N_t is the total number of assembly tasks.

Furthermore, the Garg model equations for various micromovements are as follows:

For carrying loads held against the thighs or against the waist (in kcal)

$$e_{nij} = 10^{-2} [68 + (2.54 \times WW_j) \times SW_n^2 + (4.08 \times L_n) \times SW_n^2 + (11.4 \times L_n) + 0.379 \times (L_n + WW_j) \times GS \times SW_n] t_n \quad (3.12)$$

For pushing/pulling at a bench height of 0.8 meters (in kcal/push)

$$e_{nij} = 10^{-2} (112WW_j + 1.15F + 0.505GS \times F) \quad (3.13)$$

For lateral movement of the arms at 180° in both hands (kcal/lateral movement of the arms).

$$e_{nij} = 10^{-2} (0.11WW_j + 0.726L_n) \quad (3.14)$$

For holding at arm's length in one hand (in kcal),

$$e_{nij} = 0.088L_n \times t_n \quad (3.15)$$

For pushing/pulling, at a 1.5-meter height (kcal/push),

$$e_{nij} = X(0.086 + 0.036F) \quad (3.16)$$

Carrying loads held at arm length at the sides (in one or both hands) (kcal),

$$e_{nij} = 10^{-2} [80 + 2.43 \times WW_j \times SW_n^2 + 4.63L_n \times SW_n^2 + 4.62L_n + 0.379(L_n + WW_j) \times GS \times SW_n] t_n \quad (3.17)$$

For holding at arm's length, against thighs, or at sides (both hands) (kcal),

$$e_{nij} = 0.037L_n \times t_n \quad (3.18)$$

For holding against the waist (in kcal),

$$e_{nij} = 0.062L_n \times t_n \quad (3.19)$$

where

E is the rate of metabolism (in kcal min),

e_{nij} is the energy expenditure (kcal) for carrying, walking, or holding,

WW_j is the weight of the operator (in kg),

F is the average force applied to push or pull the hands (kg),

GS represents the grade of the walking surface (%),

h_{1n} is the height from the ground to the starting point of the lifting or endpoint of the lowering micromotion,

h_{2n} is the height from the ground to the endpoint of lifting or the starting point for lowering the micromotion,

L_n represents the weight of the load (in kg),

g_j represents the gender of the operator, namely 0 for females and 1 for males,

SW_n is the speed of walking (in m/s),

X is the horizontal movement of work piece (in m), and

t_n is the time taken (in min).

The energy expenditure was calculated using the Garg model equations.

The physical limit of workers for professional applications is assumed to be 35% of the maximum energy expenditure rate, $Wmax_j$ [in W], thus providing the occupational energy expenditure (OEL_j) [in W]. Occupational energy expenditure is the power required for a worker to work for 8 h, which is the limit of the worker and is calculated using Equation (3.20).

$$OEL_j = 0.35 \times Wmax_j \quad (3.20)$$

The maximum energy expenditure rate for a worker was calculated based on the maximum oxygen consumption, VO_{2j} [l/min]. The equivalent energy at maximum oxygen consumption was assumed to be 21.25 kJ/l, as per Dalle Mura and Dini (2019). Using this value, the energy expenditure rate at maximum work is obtained using Equation (3.20).

The metabolic equivalent (MET) is a measure of the accumulated metabolic workload during an activity. The mean MET during the activity was considered to be 354.3, as the amount of physical activity was minimal in this study.

$$Wmax_j = 354.3VO_{2j} \quad (3.21)$$

The maximum oxygen consumption of a worker was calculated using the formula proposed by Schacherer et al. (1992), as per Equation (3.22).

$$VO_{2j} = \begin{cases} 66.734 - 0.315 a_j - 0.678 PFAT_j & \text{if the worker is male} \\ 58.904 - 0.356 a_j - 0.494 PFAT_j & \text{if the worker is female} \end{cases} \quad (3.22)$$

where a_j is the age of the worker and $PFAT_j$ represents the body fat percentage, which is estimated using the relationship proposed by Jackson-Pollock, as given in Equation (3.23).

$$PFAT_j = 1.61 \times BMI_j + 0.13 \times a_j - 12.1 \times g_j - 13.9 \quad (3.23)$$

where BMI_j is the body mass index and g_j is the gender of the worker. Body mass index is estimated using Equation (3.24)

$$BMI_j = \frac{WW_j}{HW_j^2} \quad (3.24)$$

Thus, the calculated energy limit is considered the physical limit of the worker, and the energy expenditure of the worker should not exceed this limit for each workstation, as represented in Equation (3.25):

$$E_j \leq 0.0143 \times OEL_j \times CT \quad (3.25)$$

where CT is cycle time (min).

In the mixed-model ALB, the product mix changes frequently in addition to the stochastic task time, thus countering a range of NP-hard problems without optimal solutions. This finding highlights the need for simpler methods. The energy expenditure of workers in the mixed-model assembly lines was calculated using the virtual average model developed by Battini et al. (2011). The weighted average energy expenditure was calculated using Equation (3.26).

$$\text{Weighted average energy} = \sum \sum p_k \times E_{ikj} \quad (3.26)$$

where p_k is the proportion of product k in the repeating sequence and E_{ikj} is the i th task's energy expenditure of product k for worker j assigned to the workstation. Thus, the energy time constraint for mixed-model ALB is

$$\sum \sum p_k \times E_{ikj} \leq 0.0143 \times OEL_j \times CT \quad (3.27)$$

Cycle-time constraint

In single-product ALB, the total task time of all the tasks assigned to a workstation should not exceed the total cycle time of the assembly line, as shown in Equation (3.28).

$$T_i = \sum_{n=1}^{Nm} t_n \quad (3.28)$$

The task time (T_i) is the sum of the execution times of all the micromovements involved in the process, as given by Equation (3.29).

$$T_i \leq \text{Cycle time (CT)} \quad (3.29)$$

In a mixed-model ALB, multiple models are assembled simultaneously; hence, the weighted average time is considered rather than

the task time. The weighted average time is the time required at a workstation to perform tasks for different products. The weighted average time at each workstation cannot exceed the required cycle time for all products. The weighted average time was estimated using Equation (3.30):

$$\text{Weighted average time} = \sum \sum p_k \times T_{ik} \quad (3.30)$$

where p_k is the proportion of product k in the repeating sequence and T_{ik} is the task time for the i th task of product k assigned to the workstation.

Therefore, the cycle-time constraint for the mixed-model ALB is given by Equation (3.31).

$$\sum \sum p_k \times T_{ik} \leq \text{Cycle time (CT)} \quad (3.31)$$

Problem instance for a single-product assembly line

In this study, a problem consisting of ten tasks and six workers was considered. The characteristics of workers available for the assembly line and their skills are listed in Table 1.

The skill level of the assigned worker should be greater than that required to perform a task at a workstation. The skill levels required for each task in the assembly lines are presented in Table 2. The micro-movements involved in each task of the considered problem instance are also listed.

The precedence constraint is that Task 1 must be completed before Task 2 and Task 9 must be completed before Task 10. To address this problem, the daily demand for a single product was 48 units with a daily working time of 8 h. Therefore, the cycle time at each workstation was computed to be 10 min. Moreover, the energy of a worker should be greater than the sum of the energies required to perform all tasks at a workstation.

3.2 | Problem instance for a multiproduct (mixed model) assembly line

In this study, the number of products in the multiproduct assembly line was set to six. The precedence relationship stated in Section 3.2 for a single-product assembly line is followed in this problem instance for a multiproduct assembly line. As previously

TABLE 1 Characteristics of workers available and their skill level for an assembly line.

Worker	Weight (kg)	Age (years)	Gender	Height (m)	Skill level	
					Single-product assembly line	Mixed model assembly line
1	75	50	Male	1.81	1	1
2	85	45	Male	1.7	2	2
3	55	30	Female	1.59	1	2
4	70	50	Female	1.65	2	2
5	70	30	Male	1.75	0	2
6	63	40	Female	1.7	0	1

TABLE 2 Skill level required for each task in the assembly lines.

Task	1	2	3	4	5	6	7	8	9	10
Micromovements	Arm lowering	Walking	Arm lifting	Carrying load against thighs or waist	Arm lifting, walking, and arm lowering	Pushing at bench height	Arm lifting	Arm lifting, lateral movement of arms (180°), and arm lowering	One hand lifting, holding at an arm's length in one hand, and arm lowering	Pushing at 1.5-m height
Skill level required	Single-product assembly line	0	1	0	1	1	0	0	2	0
	Mixed model assembly line	2	1	0	0	1	0	1	0	2

stated, the skill of the assigned worker should be greater than the skill required to perform the task at the workstation. Moreover, the worker's energy should be greater than the energy required to perform all tasks assigned to a workstation. Thus, when calculating the sum of the energies in a mixed-model assembly line, the tasks required for all six products must be considered. All tasks and their sequences were the same as those for the single-product assembly line problem instance. As mentioned above, the characteristics of the workers available for assignment to the mixed-model assembly line and their skill levels are presented in Table 1 and the skill levels required for the mixed-model assembly line are presented in Table 2. The daily demand and cycle time for each of the six products are listed in Table 3. The times required for each task for the six products in the mixed-model assembly line are presented in Table 4.

If Tasks 1 and 3 are assigned to workstation 1, then the total weighted average time for workstation 1 is calculated as follows:

$$\text{Total weighted average time} = \sum_k \sum_i (q_k \times tik) = 2.765 \text{ min}$$

The weighted average time at each workstation should be less than or equal to the required cycle time.

3.3 | Relaxation of an objective for solving the ALBP

As reported in the literature, the problem of deterioration becomes more prevalent when solving a multi-objective optimization model with an increasing number of objectives. Hence, a dimensionality-reduction approach was adopted to simplify the methodology. Thus, among the three objectives considered in the present study, the first was to minimize the number of workstations that were fixed. Four workstations were 4.

3.4 | Solution methodology proposed for an ALBP

In the present research, to solve an ALBP considering ten tasks for single- and multiproduct assembly, the following two multi-objective optimization solution approaches are proposed:

TABLE 3 Demand and cycle time for six products.

Product	Daily demand (units)	Required cycle time (min)	Demand proportion
A	20	24	0.2
B	20	24	0.2
C	15	32	0.15
D	20	24	0.2
E	10	48	0.1
F	15	32	0.15

TABLE 4 Task time of various products in a mixed model assembly line.

Task	Product task time (in min)					
	A	B	C	D	E	F
1	1.78	0.88	1.27	1.57	1.80	1.00
2	1.57	1.68	1.60	1.05	0.90	1.98
3	0.85	1.42	1.10	1.95	1.68	1.47
4	1.95	1.40	0.55	0.68	1.35	1.10
5	1.90	1.88	1.40	1.00	1.08	1.08
6	1.25	0.55	0.65	1.42	1.87	1.07
7	0.62	2.00	1.50	0.95	1.07	1.85
8	1.52	0.72	0.63	0.70	1.90	0.72
9	0.85	0.73	1.52	0.55	0.73	0.65
10	1.25	1.12	0.70	1.40	1.20	1.97

- NSGA-II and
- MOSA

Contrary to the weighted solution approaches reported in the literature, a Pareto optimality-based approach was applied in the present study to obtain nondominated solutions. This is the unique contribution of this study to the scientific literature on ALBP. In this study, the following two variants of ALB were considered:

1. Single-product ALB
2. Multiproduct ALB

In both scenarios, ten tasks were involved in an assembly operation to be carried out at four workstations. Hence, these ten tasks are to be assigned among the available workers in the four workstations, such that the total required skills can be minimized. In addition, the variance in the total energy expenditure among the workstations should be minimized; that is, the task should be distributed according to the worker's physical capability.

3.5 | Solution methodology for a single-product ALBP

This subsection illustrates the steps to be followed to solve the single-product ALBP using NSGA-II and MOSA.

The skills and micromovements required for the tasks are listed in Table 2. The daily demand for the products was 48 units, and the daily working time was 8 h. Using the available data and the proposed NSGA-II or MOSA algorithm, a Pareto front was developed to select the best task and worker sequences. The steps for obtaining the solutions are as follows:

Step 1: Calculate the energy required to perform each of the ten tasks by each of the six workers using the Garg model for a single product. The energy expenditure of each worker per task was

calculated and tabulated based on the Garg model equations explained earlier for each task and the task frequency.

Step 2: Calculate the workers' occupational energy limit (OEL) based on worker characteristics.

Step 3: Implementation of NSGA-II or MOSA

Step 4: Generation of an initial feasible population

Step 5: Calculation of objective function values

Step 6: Development of a Pareto Front

3.6 | Solution methodology for a multiproduct (mixed model) ALBP

In the mixed-model problem, six products were assembled. Hence, the assembly line is balanced so that the demand for all products is fulfilled. In this scenario, the task required to be completed for all six products was the same as that in the single-product scenario, as mentioned in Subsection 4.1. The energy available to a worker should be greater than the sum of the energies required to perform all tasks at a workstation. When calculating the sum of the energies in the mixed-model assembly line, the tasks for all six products should be considered.

3.7 | An illustrative example of the proposed solution approach using NSGA-II

The GA starts by generating an initial population of the required size. The fitness function was then calculated for all individuals in the initial population. Subsequently, based on the fitness value, individuals with the best fitness values survive to the next generation. Crossover and mutation operations are then applied to the initial population to generate a population for the next generation. This process was continued until a new population of the required size was formed.

In this study, a chromosome was designed to represent the task sequence and the worker assignment. An order-based crossover was applied to the task sequence to ensure feasible solutions, and a swap mutation was applied to the worker assignment on a chromosome.

3.7.1 | Generation of the initial feasible population in NSGA-II

Several feasible chromosomes, equal to the required population size, were randomly generated. Details about the solution approach is provided as a supplementary material. The chromosome was generated as per the flowchart shown in Figure 1. A sample scheme for a feasibly generated chromosome is presented in Supporting Information S1: Table S1 (Refer Supplementary material). Similarly, an initial feasible population of a given size was generated.

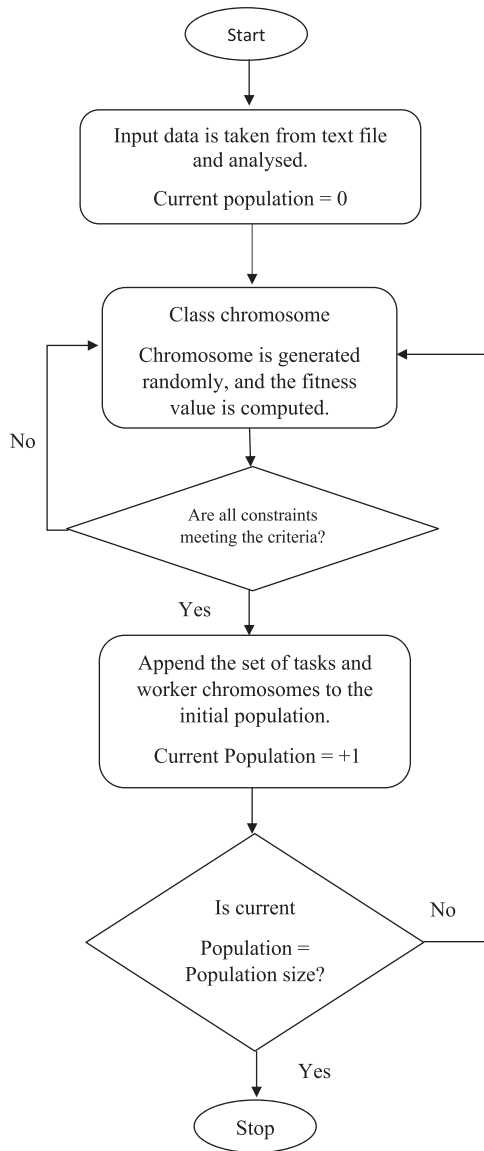


FIGURE 1 Flow chart of generating initial feasible population in NSGA-II.

3.7.2 | Calculation of objective function values

Objective 1 (O1) minimizes the number of workstations, as mentioned in Section 3.4. Although the first objective was to minimize the number of workstations and reduce the complexity of computation, the first objective was relaxed in the initial phase of the work. Therefore, the number of workstations was fixed at four, and the worker assignments and task sequences were determined using NSGA-II.

Objective 2 (O2) minimizes the total required skill. For chromosome 1, the objective function value O2 is computed as follows:

$$O2 = 1 + 1 + 2 + 1 + 1 = 6$$

Objective 3 (O3) minimized the variance in the total energy expenditure at each workstation, and the objective function value O3 was estimated as follows:

$$O3 = \sqrt{\frac{\sum_{m=1}^{wst} (n - n_m)^2}{k}} \quad (4.1)$$

where n is the mean energy gap between workstations.

wst is the number of workstations.

n_m is the energy gap of workstation m and is calculated using Equation (4.2)

$$n_m = 1 - \frac{Total E_j}{0.0143 \times OEL_j \times CT} \quad (4.2)$$

where $Total E_j$ is the sum of the energy required to complete a task by worker j at the workstation m .

OEL_j is the occupational energy limit of worker j , and CT is the cycle time in minutes.

3.7.3 | Fitness calculation for all chromosomes in the feasible population

As stated above, the novel contribution of this study is the application of the Pareto optimality approach to solve an ALBP. In the Pareto optimality approach, instead of assuming weighting for the three objectives, fitness is determined based on non-dominated solutions. A Pareto front was plotted between the values of the objective functions, O3 and O2, for all chromosomes in the population. In the plot, equally good chromosomes are represented as one front and assigned the same fitness value. Therefore, the chromosomes in front 1 had the highest fitness value, followed by the chromosomes in the next front.

3.7.4 | Generation of next-generation population

Chromosomes in the next-generation population were generated based on the previous population using operators such as elitism, crossover, and mutations.

Elitism

In elitism, the given number of chromosomes with the highest fitness is directly passed on to the next-generation population to guarantee its presence.

Crossover

To satisfy the precedence constraints given for the tasks and to ensure feasible chromosomes, an order-based crossover is applied in this study. The chromosome for the crossover was selected from the population using tournament selection. The task sequence on the chromosome was subjected to an order-based crossover. As indicated in the Supporting Information S1: Table S2, an

order-based crossover cuts the chromosomes of the two parents at two randomly chosen points. The first child chromosome is created using the head and tail portions of the first parent chromosome and by inserting the missing genes in that order, as they appear in the second parent chromosome. The same process, but the opposite, is performed to obtain the second child chromosome. This genetic operator ensures that both child chromosomes are feasible with respect to the precedence constraints. At each iteration, individuals in the current population were selected to produce offspring. In this study, order-based crossover was applied to the task chromosome. As shown in Supporting Information S1: Table S2, it works by selecting cutoff points of 2 and 8. The first offspring/child is generated by the head and tail of the first parent and the missing genes in the order in which they appear in the second parent; the opposite is true for the second offspring/child. This genetic operator was implemented to ensure that populations made only feasible solutions, which means assembly sequences respecting the precedence constraints.

4 | MUTATION

A swap mutation was applied to the worker assignment in the chromosome if the mutation probability was greater than a given value (Refer Supporting Information S1: Table S3). It is applied to a worker chromosome, and two genes are exchanged within the same individual. Specifically, this operator reverses the positions of the two workers assigned to different workstations and is implemented to ensure randomness.

This process is iterated for a given number of generations to obtain the best chromosome, comprising the best task sequence and worker assignment. For illustrative purposes, a hypothetical problem consisting of 10 tasks to be performed at four workstations in both single-product and multiproduct assembly line scenarios was considered.

4.1 | An illustrative example of the proposed solution approach using MOSA

Simulated annealing (SA) has been reported to be a reliable approach for solving single-objective optimization problems. The simulated annealing algorithm is a metaheuristic for approximately optimizing solutions in a large search space. It is mimicked by a metallurgical process that involves heating and controlling the cooling of materials to increase their crystal size and reduce their defects. In simulated annealing, when annealed slowly and sufficiently, it converged to the global optimum. However, researchers have recently focused on optimizing multiple objectives. The simultaneous optimization of several objectives is usually impossible; thus, a curve representing two objectives or a surface representing three or more objectives describes the trade-off between objectives. These curves or surfaces are known as the Pareto fronts (Smith et al., 2008). In the literature, simulated annealing has been adapted to solve multi-objective

optimization problems by combining the objectives into a single-objective function. This is known as a weighted approach. The unique contribution of the present work is that it replaces the weighted approach and applies the Pareto optimality approach, providing a set of compromising solutions for decision-makers and adopting MOSA.

In MOSA, two sets of solution sequences were initially randomly generated. Each set contained task and worker sequences. These sequences were used as seed sequences for MOSA. In the SA algorithm, poor solutions are temporarily accepted to explore the solution space. The algorithm randomly selected a solution that was close to the current solution at each step. Its quality is then measured, which determines whether to proceed with the solution or stick to it. The advantage of the simulated annealing methods is that they require a reasonable amount of time to obtain near-optimal solutions.

4.1.1 | Phase 1 of MOSA

In this phase, the first set of randomly generated solution sequences was used as the initial sequence for all search directions. The search direction was varied using two variables, x_1 and x_2 (relative to the total skill level required and the variance in energy expenditure).

In the first step, $x_1 = 0$, $x_2 = 1$, and $S = S_1$. Then, generate a uniform random number, u , in the range $[0, 1]$; if u is less than x_2 , then perturbation is employed on S , and a new sequence S' is obtained. Then, this new sequence S' (obtained by the perturbation scheme) and S are compared with respect to the energy variance objective; otherwise, (i.e., $u > x_2$), S and S' are compared to Objective 2, representing the total skill level required.

The simulated-annealing, technique-based search was performed for a predefined number of iterations, and then the search process was restarted with $x_2 = x_2 - 0.1666$, $x_1 = 1 - x_2$, and $S = S_1$. This process is continued until $x_2 < 0.5$. This search method attempts to determine as many Pareto-optimal solutions as possible within a sequence of search directions. The initial value of the direction parameter $x_1 = 0$ (or $x_2 = 0$, depending on the phase) is increased by 0.1666 in subsequent steps to divide the range $[0, 1]$ into six regions such that eight combinations of x_1 and x_2 are obtained and used to discover Pareto-optimal solutions in the search process.

4.1.2 | Phase 2 of MOSA

In the second phase, a second set of randomly generated solution sequences is given as the initial sequence for all search directions. The search direction was varied using two variables, x_1 , and x_2 (relative to the total skill level required and the variance in energy expenditure).

In the first step, $x_1 = 1$, $x_2 = 0$, and $S = S_2$. Then, generate a uniform random number, u , in the range $[0, 1]$; if u is less than x_1 , then perturbation is employed on S , and a new sequence S' is obtained. Then, this new sequence S' (obtained by the perturbation scheme) and S are compared with respect to the total skill objective; otherwise

(i.e., $u > x_1$), S and S' are compared with respect to Objective 3, representing the variance in energy expenditure.

The simulated-annealing, technique-based search was carried out for a predefined number of iterations, and then the search process was restarted with $x_1 = x_1 - 0.1666$; $x_2 = 1 - x_1$; and $S = S_1$, and this process was continued until $x_1 \leq 0.5$. In such a sequence of search directions, the search process attempts to uncover several Pareto-optimal solutions.

The parameters used for the simulated annealing technique were as follows:

- Acceptance-probability function, F
- Epoch length (number of iterations at a given temperature)
- Initial temperature, T
- Final temperature
- Cooling rate and
- Termination criterion

The acceptance-probability function for a new sequence S' is given by Equation (4.3).

$$F(S, S', T) = \exp\{-(Z(S') - Z(S)) \times 100/Z(S)/T\} \quad (4.3)$$

In the above equation, S' is the sequence obtained through the perturbation scheme, called the random-insertion perturbation scheme (RIPS), employed on S (the initial sequence), and T denotes temperature. The temperature progressively decreased from the initial positive value to the final temperature. At each step, the SA heuristic randomly selects a solution S' that is close to the current one. If the movement to S' is not economical, it is decided according to the temperature-dependent probabilities of selecting better or worse solutions. The parameters were selected such that the system ultimately moved to a lower energy level. The initial temperature was 575°C, the cooling rate was 0.9, and the final temperature was 20. The total number of iterations was 30; therefore, 30 new sequences were generated for each temperature. An illustration of this algorithm is shown in Figure 2.

5 | RESULTS

An extensive computational study was conducted using the following two solution approaches for two production scenarios:

1. NSGA-II approach
 - a. for a single-product ALBP
 - b. for a multiproduct ALBP
2. MOSA approach
 - a. for a single-product ALBP
 - b. for a multiproduct ALBP

5.1 | Application of NSGA-II for a single-product ALB

For the instance of a ten-task problem for a single-product assembly line, a solution approach based on NSGA-II was adopted. A population size of 200 and a mutation rate of 0.25 in 10 generations were considered. Using Pareto optimality, a set of equally effective, non-dominated solutions was obtained. Kindly refer to the supplementary material Tables, Supporting Information S1: Table S4a–c for the nondominated set of solutions generated by the model for the problem instance, and their corresponding objective function values are listed in Supporting Information S1: Table S5. The chromosomes were non-dominated solutions obtained using NSGA-II, and a Pareto front is depicted in Figure 3 for the single-product assembly line. This plot consists of Pareto fronts from each generation.

5.2 | Application of NSGA-II for a multiproduct (mixed model) ALB

All the parameters considered in the mixed model were the same as those in the single-product model, including the number of tasks, generations, and mutation rate. Using Pareto optimality, a set of equally good Pareto-optimal solutions is obtained. The non-dominated set of solutions generated by the model for the ten-task problem are provided in the Supplementary material as Supporting Information S1: Table S6a–c, and their corresponding objective function values are listed in Table S7. The above-cited chromosomes are non-dominated solutions obtained using NSGA-II, and a Pareto front is depicted in Figure 4 for the multiproduct (mixed model) assembly line.

After obtaining the results shown above, the task can be allocated to a workstation based on the task sequence, and the worker can be assigned to a workstation based on the worker sequence in an assembly line. From the Pareto chart for both single products, as shown in Figure 3, and multiple products, as shown in Figure 4, it can be inferred that both objective values decrease as the number of generations changes from 1 to 10. In the Pareto optimality approach, instead of obtaining only one solution, a set of Pareto-optimal solutions was obtained without assuming any weighting of the objectives. Hence, the Pareto optimality approach is considered superior to the weighted approach. This is a unique contribution of the present study to the multi-objective ALBP domain.

5.3 | Application of MOSA for a single-product ALB

For the instance of a task problem for a single-product assembly line, a solution approach based on MOSA was adopted. The non-dominated set of solutions generated by the model for the hypothetical ten-task problem are provided in the Supplementary

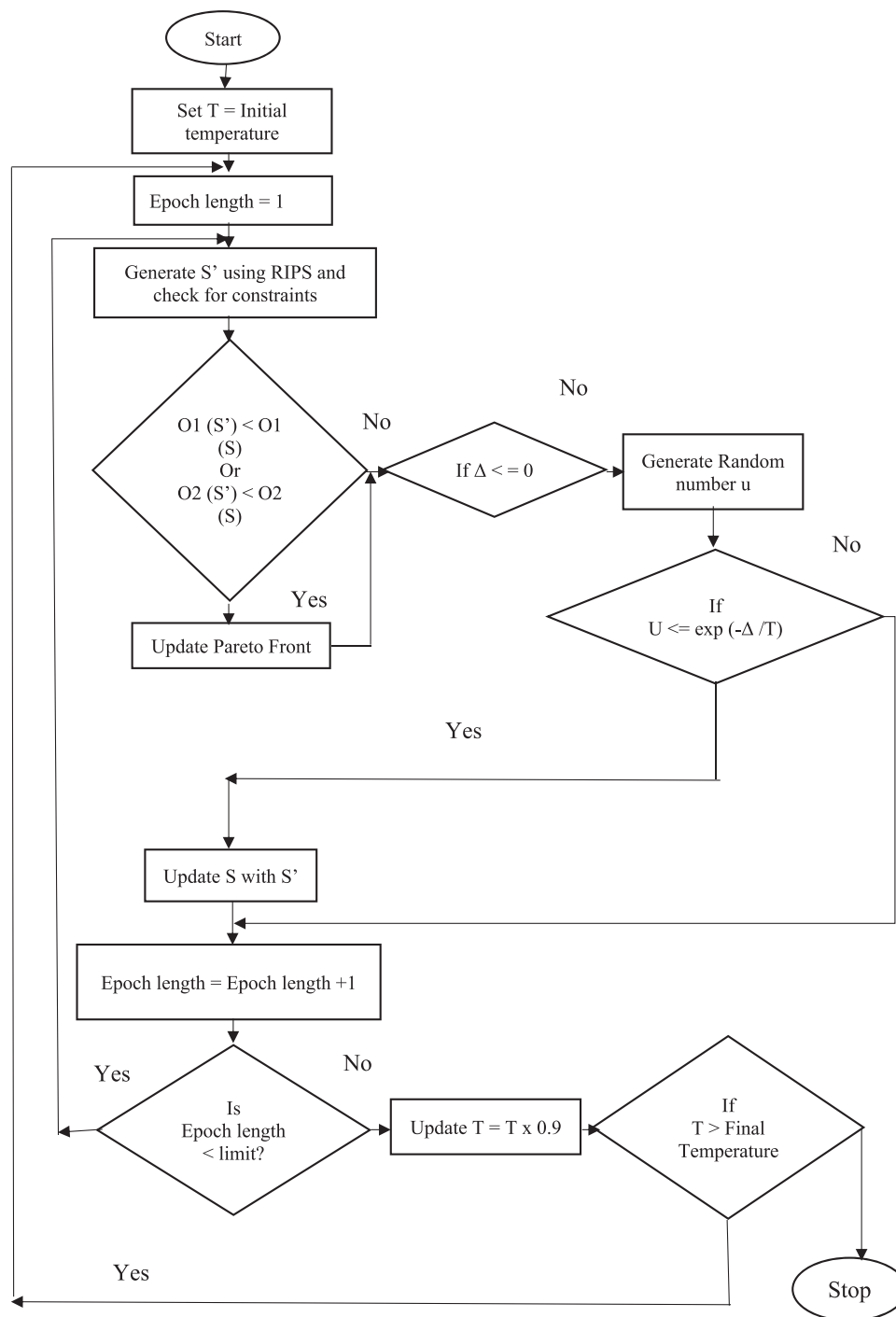


FIGURE 2 Flowchart of MOSA approach used for plotting Pareto front.

material as Supporting Information S1: Table S8a–c, and their corresponding objective function values are listed in Supporting Information S1: Table S9. The chromosomes cited above are non-dominated solutions obtained using the MOSA, and the Pareto front is depicted in Figure 5 for the single-product assembly line. This plot consisted of a Pareto front from a single generation. The Pareto front consists of non-dominated solutions with total skill level (Objective 2) and total energy variance (Objective 3) values, as shown in Figure 5.

It is inferred that the total skill objective for the best solution is obtained at 2, which is the lowest among all the generated solutions. However, its energy variance objective is 0.023, which is higher than that of many other solutions. The energy variance among the workstations that had the lowest value for the set of sequences represented in Supporting Information S1: Table S9 (Refer Supplementary Material) was 0.000138; however, its total skill objective was higher at 5. The algorithm helps to identify the set of nondominated solutions, and based on the

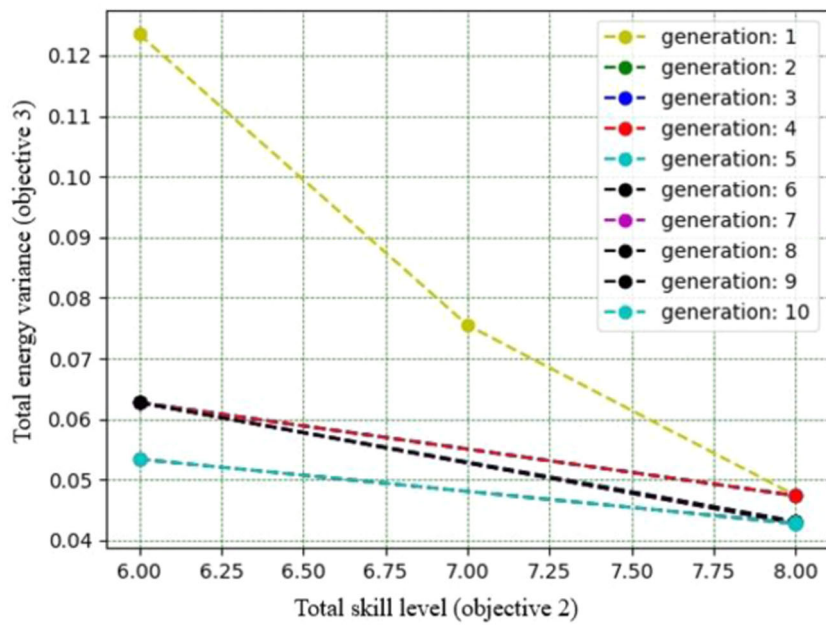


FIGURE 3 Plot of Pareto front using NSGA-II for a single-product problem instance.

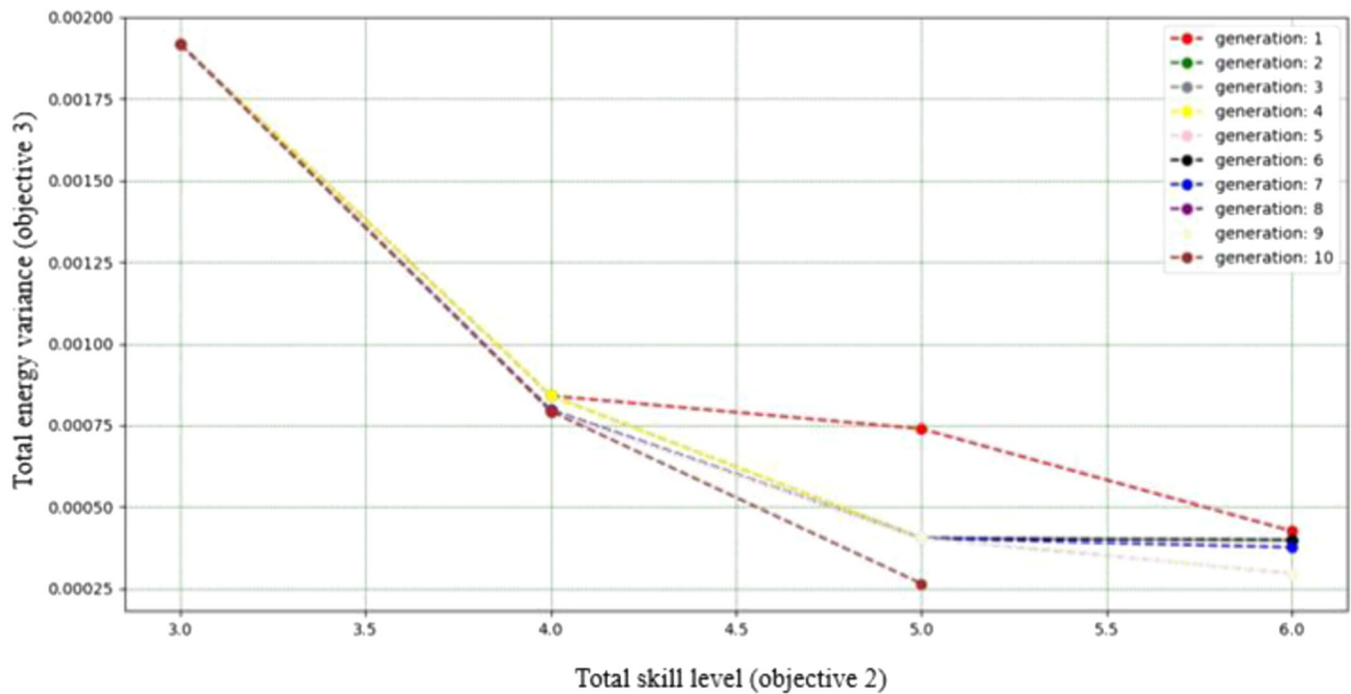


FIGURE 4 Plot of Pareto front using NSGA-II for a multiproduct problem instance.

requirements and priorities, a decision-maker can choose the best sequence for line balancing.

5.4 | Application of MOSA for a multiproduct (mixed model) ALB

For the problem instance of ten tasks for a multiproduct (mixed-model) assembly line, a solution approach based on MOSA was

adopted. The non-dominated set of solutions generated by the model for the hypothetical ten-task problem are provided in the Supplementary material as Supporting Information S1: Table S10a–d, and their corresponding objective function values are listed in Supporting Information S1: Table S11. It can be observed from Supporting Information S1: Table S11 (Refer Supplementary material) that the lowest value of the total skill level, that is, for Objective 2, is 2 for a particular set of tasks, whereas the lowest value of the energy variance (i.e., for Objective 3) is in non-dominated Solution 4, as

FIGURE 5 Plot of pareto front using MOSA for a single-product problem instance.

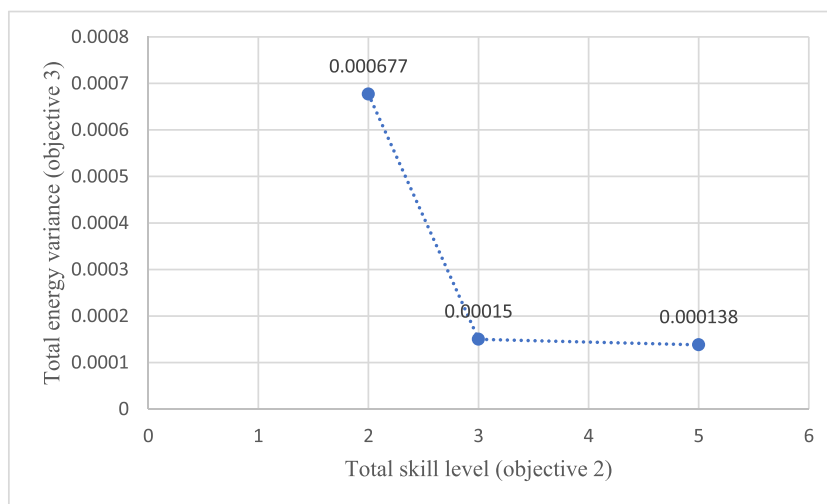
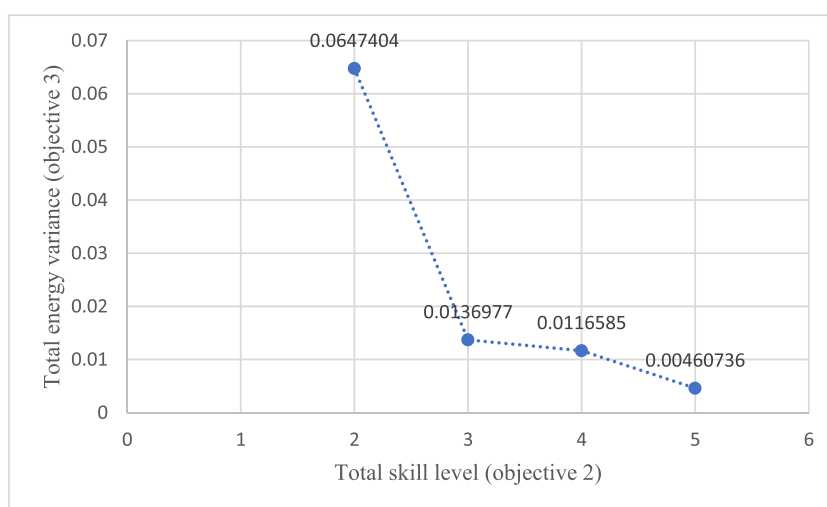


FIGURE 6 Plot of pareto front using MOSA for a multiproduct problem instance.



shown in Supporting Information S1: Table S11 (Refer Supplementary material). A Pareto front is plotted, which consists of nondominated solutions with the total skill level (Objective 2) and total energy variance (Objective 3) values, as shown in Figure 6.

6 | COMPARATIVE ANALYSES OF THE PROPOSED SOLUTION APPROACH NSGA-II AND MOSA

For any solution approach, the computational time, or runtime, is crucial for indicating its efficiency. It has been reported that, for any combinatorial-type problem, the runtime increases exponentially as the size of the problem increases. In this computational study, the design of experiments was applied to identify the effect of input factors on the runtime of a solution approach in a multiproduct (mixed model) assembly line.

In this study, five critical factors were identified that varied between the two levels, as shown in Table 5. Precedence constraints were

expressed as a percentage of the number of tasks and were set at 10% and 20%. The skill levels were rated on 3- and 5-point scales. Similarly, the number of workstations was set to four and five, the number of tasks to 10 and 12, and the number of product models to six and eight.

For a comparative analysis of the proposed solution approaches in terms of computational time and runtime, an elaborate computational experiment was performed. The model was replicated three times for each run of the experiment, and the respective run times were recorded. Thus, $2^5 \times 3$ trials were performed. To investigate the influence of various factors, such as precedence constraints, skill levels, number of workstations, number of tasks, and number of products, on the two proposed solution approaches, each was applied to a set of 32 hypothetically generated problem instances at different combinations of the two levels of factors, as listed in Table 5. For each problem instance, both of the proposed approaches were applied and run three times. The computational time or runtime required for both of the proposed solution approaches in each iteration is listed in Table 6.

TABLE 5 Factors and its levels identified for the design of experiments.

Factors	Level 1	Level 2
Precedence constraints (% of task)	10	20
Scale adopted for rating skill levels	3	5
Number of workstations	4	5
Number of tasks	10	12
Number of models	6	8

Considering the three iterations, it was found that the average runtime for NSGA-II was 37.484 s, and that for MOSA was 10.012 s. Therefore, it can be concluded that the MOSA computationally outperformed the NSGA-II.

6.1 | Performance analyses

Furthermore, the Pareto charts of the standardized effects of the factors listed in Table 5 on the proposed solution approaches, NSGA-II and MOSA, are depicted in the Supplementary material as Supporting Information S1: Figure S1a,b, respectively. With reference to NSGA-II, it was observed that the number of tasks had the greatest influence on the runtime of the solution approach, followed by the percentage of precedence constraints and the interaction effect of both. Therefore, it is not advisable to apply NSGA-II as a solution as the number of tasks increases. Similarly, in situations with a higher precedence relative to the number of tasks, NSGA-II requires more computational time.

Moreover, with reference to MOSA, it is observed that the interaction effect of the number of workstations and number of tasks has the greatest influence on runtime, followed by the number of workstations, and then by the interaction effect of skill level, number of tasks, and number of products/models. However, the MOSA performed relatively better in both production scenarios. Furthermore, to investigate the effect of individual factors on the runtime of the solution approaches, NSGA-II and MOSA, a main effects plot was analyzed, as depicted in the Supplementary material as Supporting Information S1: Figure S2a,b, respectively.

From Supporting Information S1: Figure S2a (Refer Supplementary material), it can be inferred that the number of tasks has the greatest influence on the runtime of NSGA-II. Similarly, with reference to Supporting Information S1: Figure S2b (Refer Supplementary material), the number of workstations has the greatest influence on the runtime of the MOSA. The runtime of the solution approach increases when the various factors considered increase, except for the number of products in NSGA-II and the skill levels in MOSA.

The interaction plot of the factors helps to interpret the main effects between the lines. Therefore, the interaction plots of the

factors considered with respect to the solution approaches of NSGA-II and MOSA are shown in the Supplementary material as Supporting Information S1: Figure S3a,b, respectively. When analyzing these interaction plots, if the lines are not parallel to the x axis, it can be inferred that there is a main effect between the factors considered; that is, across all factor levels, the response mean is not the same. The steeper the slope of the line, the greater the magnitude of the main effect. According to Supporting Information S1: Figure S3a,b (Refer Supplementary material), a greater magnitude of the main effects is more evident in the case of the MOSA than in the NSGA-II. This proves that the solution approach of the MOSA performs better than NSGA-II in solving a multi-objective ALBP.

The results illustrate how the algorithm simultaneously minimizes the number of skilled personnel and energy load variance, thereby achieving the proposed objectives. The algorithm was tested using real-time data obtained from the assembly experiments. The algorithm helps minimize the number of workstations and simultaneously minimizes the number of skilled workers and the energy variance among workstations. Workers are less strained because their physical capabilities are considered when assigning tasks to the workstations. This maximizes the productivity of the firm and distributes the workload equally among the workstations.

7 | DISCUSSION

A multi-objective ALBP is addressed in this work by proposing NSGA-II and MOSA approaches to minimize the overall skill level required and variance in workers' energy expenditure across workstations. Mehta (2016) reported the need to integrate both the physical and cognitive aspects of ergonomics, as most of the work environment is involved. However, the integration of these aspects is limited in the literature.

The approach described in this research combines physical (e.g., energy expenditure) and cognitive ergonomics (e.g., worker skill). Solving these objectives of the mathematical model contributes to an improved workstation design by evaluating tasks to make them compatible with people's needs, abilities, and limitations. Furthermore, a comparative analysis of NSGA-II and MOSA was conducted in single- and multiproduct production scenarios to identify the dominant algorithm. The computational analysis indicated that the runtime performance of MOSA was 73.287% better than that of NSGA-II; therefore, MOSA outperformed NSGA-II. In summary, this research helps improve the productivity and efficiency of workers with proper planning and allocation by solving the ALBP.

7.1 | Key findings

The proposed approach will aid in identifying worker skills to determine the right individuals to allocate to a particular workstation, considering the

TABLE 6 Experimental design and the observed runtime for different combinations for three iterations.

Precedence constraints (%)	Skill level (scale)	No. of workstations	No. of tasks	No. of products	Runtime (s)					
					MOSA			NSGA-II		
					Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
-1	-1	-1	-1	-1	6.67	7.31	5.92	11.312	10.584	10.205
-1	-1	-1	-1	1	5.62	8.92	5.43	10.839	10.732	10.899
-1	-1	-1	1	-1	9.76	8.97	10.10	20.943	29.198	17.108
-1	-1	-1	1	1	10.58	9.89	10.27	22.104	21.654	21.461
-1	-1	1	-1	-1	12.59	11.19	10.28	13.260	12.140	12.418
-1	-1	1	-1	1	21.42	13.13	17.94	12.409	12.836	12.967
-1	-1	1	1	-1	12.43	9.38	8.27	21.34	23.346	25.242
-1	-1	1	1	1	5.48	5.58	6.80	27.464	29.296	27.915
-1	1	-1	-1	-1	6.98	5.93	6.12	12.166	11.930	11.967
-1	1	-1	-1	1	3.71	5.25	5.40	15.848	14.958	17.824
-1	1	-1	1	-1	8.04	8.72	4.08	22.392	25.454	25.029
-1	1	-1	1	1	13.98	14.94	18.54	31.119	38.497	41.626
-1	1	1	-1	-1	13.33	9.73	11.10	20.206	15.693	17.336
-1	1	1	-1	1	14.61	9.19	11.65	16.829	19.606	17.666
-1	1	1	1	-1	7.63	7.95	10.71	48.286	44.414	44.263
-1	1	1	1	1	7.99	8.36	9.43	44.595	53.620	47.770
1	-1	-1	-1	-1	5.45	6.85	5.97	15.037	21.280	20.128
1	-1	-1	-1	1	5.79	3.66	5.29	14.217	14.509	13.848
1	-1	-1	1	-1	11.00	10.51	7.81	45.700	51.753	56.784
1	-1	-1	1	1	9.99	12.6	9.79	44.248	43.517	49.422
1	-1	1	-1	-1	9.28	8.97	9.15	16.863	19.662	17.618
1	-1	1	-1	1	19.14	17.81	17.98	16.636	18.328	17.519
1	-1	1	1	-1	11.47	9.20	9.32	68.988	89.020	77.791
1	-1	1	1	1	10.12	12.40	14.14	79.663	75.415	78.221
1	1	-1	-1	-1	5.51	6.41	6.18	16.251	16.520	14.600
1	1	-1	-1	1	4.88	6.88	5.75	21.682	19.457	19.586
1	1	-1	1	-1	5.68	9.72	5.90	56.867	57.499	60.510
1	1	-1	1	1	16.40	14.11	15.34	69.194	78.047	89.787
1	1	1	-1	-1	12.72	15.11	15.37	24.668	24.341	25.762
1	1	1	-1	1	11.36	10.82	13.83	26.905	24.155	27.257
1	1	1	1	-1	13.98	9.95	10.64	135.620	134.188	143.964
1	1	1	1	1	14.51	14.18	15.02	138.817	162.264	157.631
Average runtime					10.013			37.484		

complexity of the task and also reduces the cognitive workload. In the existing literature, researchers typically employ a weighted-sum approach (Dalle Mura & Dini, 2019) to address multi-objective problems in line balancing. However, the drawback of the weighted-sum method is its reliance on the assigned weights, where even a small variation can

significantly alter the objective function values. This study addresses this drawback by utilizing the Pareto optimality method. It contributes to determining the optimal workplace arrangement by examining factors such as workstations, skills, and energy expenditure through an iterative process aimed at identifying the most efficient and ergonomic layout.

Overall, this study is an effective approach to the design of ergonomic processes, as it allows the production units to optimize multiple objectives simultaneously, leading to a more efficient and ergonomic assembly line design.

7.2 | Comparison with existing literature

The ergonomic aspects have gained increasing attention in practice and research. We concentrate on the following three main trends in the existing literature related to ergonomics in ALBP. (i) The ergonomic risk(s) of a workplace are assessed, depending on the assigned tasks, by adopting an existing method that is limited by a predefined maximum ergonomic risk value. This methodology has been previously reported by Kara et al. (2014), Otto and Scholl (2011), and Sternatz (2014). (ii) To enhance the ergonomic situation, an ergonomic term can be included in the objective function, as performed by Battini et al. (2016) and Ozdemir et al. (2021). (iii) Moreover, ergonomic aspects are combined with other practical aspects, such as mixed-model production and parallel and/or U-lines, as adopted by Mokhtarzadeh et al. (2021) and Sgarbossa et al. (2016), Tiacci and Mimmi (2018); worker assignment as done by Akyol and Baykasoglu (2019), Zhang et al. (2020c); space constraints by Bautista et al. (2016); and part feeding as in Battini et al. (2017) and Bortolini et al. (2017).

Of the three aforementioned trends, the second and third bear more resemblance to the present research. In this study, the ergonomic situation is improved by including ergonomic terms in the objective function of an optimization model. Mixed-model ALB with a worker assignment based on skills is also performed as per the third category listed above.

Samouei and Ashayeri (2019) solved a multiproduct ALBP in a straight assembly line layout, whereas in this study, both single- and multiproduct ALBP were considered for a straight assembly line layout. For small-scale production units, both single- and multiproduct ALBP must be solved for better results, as there will always be small subassembly lines where only a single product is being manufactured, which has to be linked to the main assembly line. Çil et al. (2020) solved a multiproduct ALBP problem considering a single objective, whereas this work has three-objective functions in a straight assembly line layout. Dalle Mura and Dini (2019) used a weighted-sum approach to address multi-objective ALBP. The drawback of this approach is already mentioned above and this study mitigates it by utilizing the Pareto optimality method.

Otto and Battaia (2017) provided a recent survey on research focusing on the integration of ergonomic aspects into ALB and job rotation as another lever to reduce ergonomic stress over time. Job rotation might also affect the workers' biological cycle, and when it comes to skilled-aged workers, job rotation is not recommended. This study emphasizes skills instead of job rotation, and thus, worker allocation improves productivity in the assembly line. As per Boysen et al. (2022), many alternative approaches exist depending on the focused stress factor (e.g.,

repetitiveness, static postures, unhealthy positions, required forces, and energy expenditures) and suitable methods for risk assessment. One of these, energy expenditure, was also included in this study.

7.3 | Practical implications

The different inter- and cross-disciplinary nature of this study allows a wide range of issues in human resource management to be resolved. Examples of specific subject areas of interest include selection training and education, labor-management relationships, human skills development, and knowledge-enhancing technologies.

7.3.1 | Diesel pump manufacturing unit

This study considers the assembly line of a diesel pump manufacturing unit in India. As explained above, a wide variety of small parts and end products exist. Manufacturing units experienced a decline in productivity, and it is essential to identify the reason for this decline. However, breaking down the finished product into subparts and identifying the root cause are time-consuming for the operations team. After using scientific techniques, such as the Shainin method, the operation team identified issues with the barrel and plunger. This study assessed the assembly lines for barrels and plungers, considering six types of barrel-plunger assemblies. The balls inside the assembly were interchanged, and this was identified as an issue. In one product, it was assumed to be a steel ball, and in the other, it was a ceramic ball. This problem has been studied, and it has been found that workers require proper cognitive abilities. In addition, some workers could not distinguish between the colors or weights of the balls. Thus, in the present study, some inputs were obtained from a previous study by Giridhar and Panicker (2023) on workers' cognitive skills to solve the ALBP. Kindly refer to the Supporting Information S1: Figure S4 as shown in the Supplementary material for a comparison between the existing work by Giridhar and Panicker (2023) and the present work. Using this intervention, the results from ALBP increased the first-pass yield from 55% to 89%, which helped reduce rework. This underscores the practical implications of the study. The proposed allocation method can be employed in special processing industries that use stainless steel and carbon steel to avoid weld defects and reduce material waste.

The methodology employed in this study was applied to all the production units with certain modifications. The challenge in this process is to identify a problem in an actual scenario and use it to solve it. In most cases, this problem can be resolved using experienced personnel. The problem must be converted from a subjective form into a mathematical formulation, referred to as the ALBP. One of the significant limitations of the present study is that, in each industry, the ALBP formulation is different, and the selection of the right method is crucial after defining the problem.

7.3.2 | Random worker allocation in the production units of small-scale and large-scale industries

In manufacturing industries, the random assignment of workers to certain tasks leads to inefficiency, resulting in a waste of time, resources, and the delivery of finished products to customers. The user's perceived level of mental effort, known as the cognitive workload, is influenced by various factors, particularly task load and task design. In general, the greater the skill, the lower the cognitive workload. However, there are certain exemptions when considering age, gender, etc., which are separate topics for discussion. The main reasons for the random allocation of workers by the employer are lack of skilled workers, lack of knowledge of ALB, cost constraints, space constraints, workers leaving the company for better incentives and recruiting temporary workers (on a contract basis) to reduce costs who are not skilled.

Although random allocation cannot be solved easily, the present study suggests that employers must determine the skills of workers and assign them to appropriate workstations. The proposed approach, in turn, lowers training costs and enables the training of suitable workers. Appropriate training equips individuals with skills that can be leveraged to improve their employers' overall output. This study contributes to small- and medium-sized enterprises that can apply the proposed method to their units. Moreover, according to the recommendations of ISO 26000 standards, improving wellbeing at work is a core component of social responsibility. Therefore, firms must optimize their workload in assembly lines by considering ergonomics.

7.3.3 | How ALB improves ergonomics

ALB is used to reduce waste and increase the efficiency of an assembly line. When an assembly line is balanced, each workstation is designed to have an equal workload, and each worker performs the same amount of work. Hence, this prevents the overburdening of any worker and reduces the physical stress. ALB can improve worker posture by designing workstations that accommodate different worker heights and postures. Workload balancing also helps to minimize repetitive motion and reduces the likelihood of injuries, such as tendinitis and carpal tunnel syndrome. This improves the worker satisfaction, thereby resulting in a more enjoyable work experience. Balancing the assembly line renders the process more efficient, reducing wasted motion and downtime which reduces productivity.

Efficient assembly lines require less lifting, stooping, bending, and twisting, which leads to fewer musculoskeletal disorders and significant ergonomic improvements. In conclusion, ALB can significantly improve ergonomics and result in a safer and healthier environment for workers.

India is believed to have the potential to become a Global Manufacturing Hub, and by 2030, it is expected to add more than US \$500 billion annually to the global economy. In India, the

manufacturing sector has reportedly provided employment to the tune of 35.6 million in 2023. Therefore, this study can serve as a guide for small- and medium-scale industries. This study can also be applied to large production units, provided that they start by splitting the large production units into small workstations and then expand the study to the whole unit.

7.4 | Study limitations

Only an expert in this area, who has knowledge of ALBP and ergonomics, can extract the exact data from an industrial environment and solve the problem. This is a time-consuming process that requires significant attention to obtain valid and reliable data. With regard to ergonomics, anthropometric data vary for different nationalities, and their energy expenditures vary based on the tasks that lead to a change in the results. Thus, with the methodology in this study, the modeling part can be performed, but a variation in the results is expected. The influence of ongoing technological developments such as digitization, autonomous robots, and Internet-of-Things (IoT) technology and their adaptation to ALBP are not considered in this study. The varying ALBP models that are required for the different phases of the lifecycle of an assembly line and the products it manufactures were not considered in this study. The present study mainly focused on the initial line balance. Combined models that integrate the ALBP with related decision tasks are also a limitation of this study.

7.5 | Areas for further research

In the future, we will consider job rotation when allocating tasks to the workers. Similarly, the application of collaborative robots should be considered for use in assembly lines to reduce energy expenditure and enhance worker ergonomics. The collaborative (human-robot collaboration) ALBP (C-ALBP) can be performed with automated assembly lines to identify the feasibility of expansion to improve flexibility and efficiency in assembly systems.

As a main future research topic, it is proposed to carry out an accurate analysis in terms of the obtained ALB between workstations by trying different types of ALBP. In addition, the number of tasks, product variety, experimental parameters, and anthropometric data can be varied. Furthermore, when we consider a task requiring a greater physical workload, the energy expenditure will vary, thereby influencing the results. This changed the number of iterations required to obtain a near-optimal solution.

Some areas of future research include adapting the problem for mixed-model assembly lines and different line configurations, considering the learning effect of workers, addressing a production environment in which task times are stochastic, and optimizing different performance measurements. In addition, the problem can be

solved using different heuristic and meta-heuristic methods, and comparisons can be made between them.

7.6 | Ergonomics improvement in assembly lines

Ergonomic improvement in assembly lines is crucial for the safety and wellbeing of workers as well as the overall efficiency of the assembly process. Some strategies for improving ergonomics in assembly lines are task analysis, workstation design, material handling, automation and training.

Analyze the tasks that workers perform and identify potential ergonomic hazards. This can help determine adjustments to work methods, equipment, and workstation design. Workstation design plays a significant role in assembly line ergonomics. Workstations should be designed to be adjustable and to accommodate different user heights and postures. This can involve adjusting the height of the equipment, using padded flooring, or providing appropriate lighting. Material handling is an integral part of the assembly line. The design of material handling equipment should be based on ergonomic principles with appropriate handling guidelines, techniques, and tools.

Automation can reduce manual labor and improve ergonomics in assembly lines. Automating repetitive and physically demanding tasks can reduce the risk of musculoskeletal disorders, lower the physical stress on workers, and allow them to focus on more complex tasks. Providing ergonomic training to workers can increase their awareness of potential hazards and assist in avoiding them. Additionally, workers can learn how to adjust their work posture and take breaks as needed, which can significantly reduce the risk of ergonomic injury. Overall, ergonomic improvements in assembly lines can lead to significant benefits, including a reduced risk of injuries, improved productivity, and enhanced worker satisfaction.

8 | CONCLUSIONS

The application of ergonomic aspects in assembly lines has received little attention in the literature, although considering these aspects improves productivity and minimizes ergonomic issues in assembly lines. This study focused on a multi-objective ALBP with three objectives: minimization of the number of workstations, total skill-level requirements of workers, and variance in the energy expenditure of workers among workstations. A literature review verified that many researchers have extensively investigated multi-objective ALBPs using the weighted-sum approach without considering the ergonomic aspects. To address this gap in the literature, this study proposes an approach that considers two algorithms, namely, NSGA-II and MOSA, by applying Pareto optimality.

A comparative analysis of the NSGA-II and MOSA in single- and multiproduct production scenarios was conducted. Furthermore, a computational study with varying factors was performed to identify the dominant algorithm that can solve the multi-objective ALBPs, and

the results demonstrated that the computational efficiency of MOSA was 73.287% better than that of NSGA-II.

The proposed approach minimizes the number of workstations, total skill level required for workers on the line, and energy load variance among workstations. The energy load was evaluated by comparing the energy expenditure while performing tasks with the worker's physical energy limit, which was calculated based on anthropometric and physiological factors.

The results indicated that the proposed approach could assign workers and operations to workstations while reducing the energy load variance under all conditions while considering skill constraints and energy expenditure. Even if workers have a high workload or are unable to perform assigned tasks, the algorithms for solving line-balancing problems that ignore ergonomics may generate solutions that minimize the number of workstations in an assembly line. Consequently, the recommended algorithms are well-suited for industries in which assembly activities necessitate dexterity and manual skills gained through experience. In such production units, skilled workers are valuable assets, like any other resource. The recommended algorithms can be used to design assembly plans based on specific individuals available to the organization, thereby preventing workers from being overloaded and stressed.

ACKNOWLEDGMENTS

No funding received.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare that are relevant to the content of this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Giridhar, M. P., Panicker, V. V. (2024). Effect of ergonomic aspects on single- and multiproduct assembly-line balancing problems. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 34, 491–515. <https://doi.org/10.1002/hfm.21046>